NIKHIL REDDY LEVAKU

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PROFESSIONAL SUMMARY

DevOps-focused engineer who provisions, automates, containerizes, and monitors multi-service applications across **AWS**, **Kubernetes**, and **Linux** environments. Built **KubeHeal**, an autonomous AI-powered Kubernetes SRE operator with automated self-healing and **Helm 3** packaging, alongside **Jenkins** CI/CD pipelines and modular **Terraform** (IaC). Backed by production backend API engineering in Core Java, Spring Boot, and PostgreSQL at DeepLure Research, bridging infrastructure and code.

TECHNICAL SKILLS

Cloud & Infrastructure	**AWS** (EC2, VPC, Subnets, Route Tables, IGW, Security Groups, S3, CloudWatch, ALB, Auto Scaling), **Linux** Administration, Bash/Shell Scripting
Container Orchestration	**Kubernetes** (Pods, Deployments, ReplicaSets, Services, Ingress, ConfigMaps, Secrets), **Helm 3**, **Docker**, Docker Compose
IaC & Configuration	**Terraform** (IaC Modules, State, Variables/Outputs, tfvars), **Ansible**, Chef
CI/CD & DevSecOps	**Jenkins** (Controller-Agent, Declarative/Scripted Pipelines, Jenkinsfile), **GitHub Actions** (CI), SonarQube, SonarScanner
Observability	**Prometheus** (15s Scrape, PromQL), **Grafana**, Graylog, AWS CloudWatch, Actuator, Micrometer
Backend & Systems	Java 21, Spring Boot 3, REST APIs, PostgreSQL, Redis, Apache Kafka, Nginx Reverse Proxy, Git

PROFESSIONAL EXPERIENCE

DeepLure Research Pvt Ltd — Backend & Cloud Engineer **May 2026 – Aug 2026**

Java · Spring Boot · PostgreSQL · Redis · Docker · GitHub Actions · Graylog · AWS S3 · OVH Object Storage

- Analyzed business requirements and delivered **100+ production REST APIs** in Spring Boot across B2B marketplace and on-demand platforms, building configurable backend components.
- Containerized Spring Boot microservices with **Docker** for reproducible runtime environments.
- Built **GitHub Actions** CI workflows for testing, compilation, and container image builds.
- Diagnosed and resolved production defects across API, database, and cache layers using centralized **Graylog** log aggregation.
- Created **AWS S3** and **OVH** object-storage buckets for application media storage.
- Engineered **Redis-backed asynchronous workers** for notifications and booking expiry; reduced database load **~40%** through Redis caching and query optimization.
- Implemented **JWT**/RBAC security, OTP verification, and **Razorpay** payments with idempotency keys and concurrency locking to prevent double-charges.
- Modeled normalized **PostgreSQL** schemas with Hibernate/JPA and Flyway; aligned API contracts with frontend teams via Swagger/OpenAPI; fixed defects in Agile sprints.

SmartBridge Educational Services (APSCHE) — Data Analytics Intern **May 2025 – Jul 2025**

- Cleaned sales datasets with SQL joins and aggregates and built 3 Tableau dashboards for production and inventory review.

PROJECTS

KubeHeal — Autonomous AI-Powered Kubernetes SRE Operator **Aug 2026 – Present**

Kubernetes · Helm 3 · Docker · Jenkins (CI/CD) · Gemini API · Prometheus · Grafana | GitHub

- Engineered an autonomous Kubernetes SRE operator that monitors cluster event streams in real time using client APIs to detect container failures (CrashLoopBackOff, OOMKilled, Error).
- Integrated an AI triage engine (**Gemini API**) that analyzes crash telemetry and logs to classify incidents into auto-fixable glitches vs complex application code defects.
- Automated self-healing via the Kubernetes API: graceful pod re-creation and deployment rollbacks for basic failures, while blocking recursive reboot storms and generating structured Root Cause Analysis (RCA) reports for code bugs.
- Packaged the operator as a **Helm 3** chart with least-privilege RBAC (ClusterRole, ServiceAccount) and automated deployment through a **Jenkins** Declarative CI/CD pipeline.

Flight Booking System — Distributed Microservices Platform **Aug 2025 – Jan 2026**

Terraform (IaC) · AWS (VPC, EC2) · Docker Compose · Nginx · Spring Boot · PostgreSQL · Kafka | GitHub

- Refactored a monolithic backend into 4 decoupled microservices (Auth, Flight, Booking, Notification) with per-service PostgreSQL databases and versioned **Flyway** migrations.
- Provisioned AWS infrastructure with modular **Terraform** (IaC) — custom VPC, public subnet, IGW, route tables, security groups, and an EC2 instance with user-data bootstrapping — parameterized using variables, tfvars, and local modules.
- Orchestrated the full stack with **Docker Compose** behind an **Nginx** reverse proxy (:80) with path-based routing, **Kafka** event streaming, and row-level pessimistic locking (PESSIMISTIC_WRITE) to prevent overbooking.

EDUCATION

Sri Venkateswara College of Engineering, Tirupati **2022 – May 2026**

B.Tech, Electronics & Communication Engineering CGPA: 8.5 / 10

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate — Oracle (Sep 2025) · **AWS** Cloud Practitioner (CLF-C02) Official Practice Exam — AWS Training & Certification (Nov 2025) · **DevOps** Basics for Everyone — IBM (May 2024)